CoreS3

SKU:K128





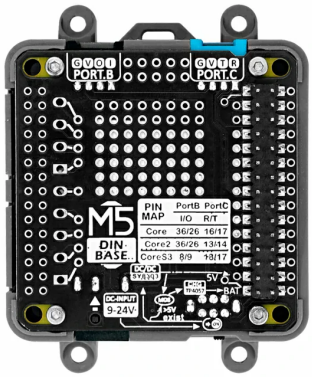


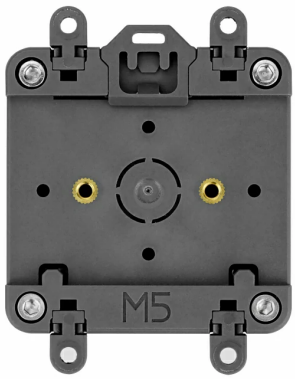














Description

CoreS3 is the third-generation main controller of the M5Stack development kit series, based on the **ESP32-S3** solution featuring a dual-core Xtensa LX7 processor running at 240 MHz with built-in Wi-Fi. It integrates 16MB Flash and 8MB PSRAM. Programs can be downloaded through the USB Type-C port, and **OTG & CDC** functions are supported for connecting USB devices and flashing firmware.

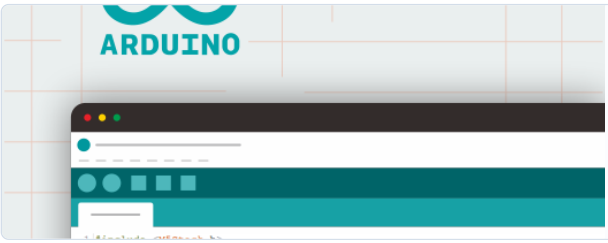
The front houses a 2.0-inch capacitive-touch IPS panel protected by high-strength glass. Below the screen is a 0.3MP GC0308 camera paired with an LTR-553ALS-WA proximity sensor. Power is managed by an AXP2101 PMU and four power-path control circuits, delivering a low-power design. Onboard sensors include a six-axis IMU BMI270 and a magnetometer BMM150. Additional peripherals comprise a microSD slot and a BM8563 RTC, enabling precise time-keeping plus sleep/timed-wake-up functions.

For audio, a 16-bit I2S amplifier AW88298 drives the built-in 1W speaker, while an ES7210 audio codec provides dual-microphone input. Independent POWER and RESET (RST) buttons are located on the side; long-pressing the RESET button enters download mode via a self-built delay circuit.

The kit ships with a DinBase for DIN-rail, wall-mount or screw mounting. It can be powered by an external DC 12V (9 ~ 24V) supply or an internal 500mAh battery. Multiple proto areas on the DinBase ease DIY expansion.

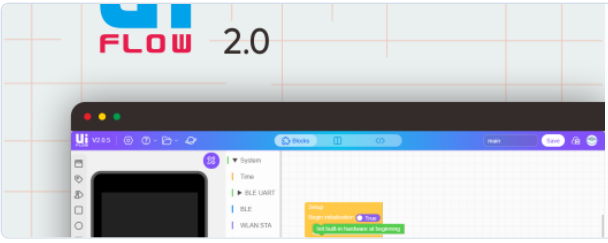
Ideal for IoT development, DIY projects, smart-home control systems and industrial automation control systems.

Tutorial



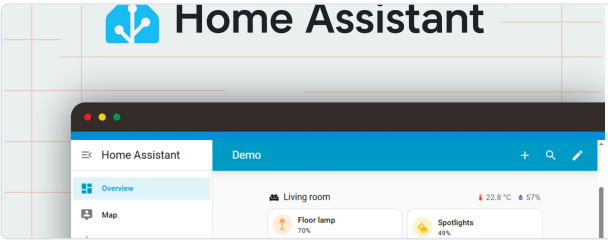
Arduino IDE

This tutorial explains how to program and control CoreS3 using Arduino IDE.



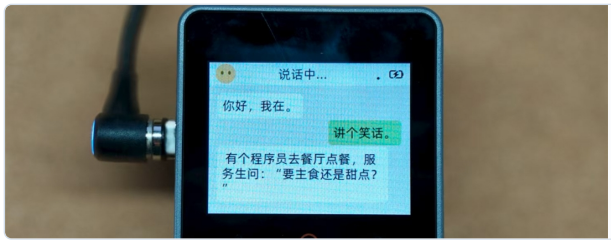
UiFlow2

This tutorial shows how to control CoreS3 via the UiFlow2 graphical programming platform.



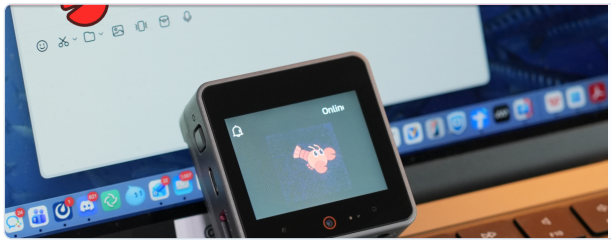
Home Assistant

This tutorial describes how to connect CoreS3 to Home Assistant.



CoreS3 Xiaoling Voice Assistant

This tutorial will introduce how to use the CoreS3 controller to burn the Xiaoling Voice Assistant firmware via M5Burner and build a personal voice assistant application.



CoreS3 ESP-Claw Firmware Flashing

This tutorial explains how to flash the ESP-Claw firmware onto the CoreS3, enabling users to quickly configure their CoreS3 as an intelligent terminal capable of AI interaction, hardware programming, and automation control.

Features

- ESP32-based development, Wi-Fi, 16 MB Flash, 8 MB PSRAM
- Built-in camera, proximity sensor, speaker, power-indicator LED, RTC, I2S amplifier, dual microphones, capacitive touch screen, power button, reset button, gyroscope
- microSD slot
- High-strength glass panel
- Supports OTG & CDC functions
- AXP2101 power management, low-power design
- Development Platform
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 × CoreS3
- 1 × DinBase
- 1 × Hex Key L-Shape 2.5 mm (For M3 Screw)
- 4 × Screw Buckle
- 1 × Rail-mount Base Buckle

Applications

- IoT development
- DIY project development
- Smart-home control systems
- Industrial automation control systems

Specifications

Specification	Parameter
SoC	ESP32-S3 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz
Flash	16MB
PSRAM	8MB Quad
Wi-Fi	2.4 GHz Wi-Fi
Touch IPS LCD	2.0" @320 × 240 ILI9342C
Camera	GC0308 0.3MP
Proximity Sensor	LTR-553ALS-WA
Power Management	AXP2101
6-axis IMU	BMI270
3-axis Magnetometer	BMM150
RTC	BM8563
Speaker	16-bit I2S amplifier AW88298 @1W
Audio Codec	ES7210, dual-microphone input
Product Size	Whole set (CoreS3+DinBase): 69.0 × 54.0 × 31.5mm Main unit (CoreS3): 54.0 × 54.0 × 15.5mm
Product Weight	72.7g
Package Size	105.6 × 66.0 × 35.3mm
Gross Weight	101.8g

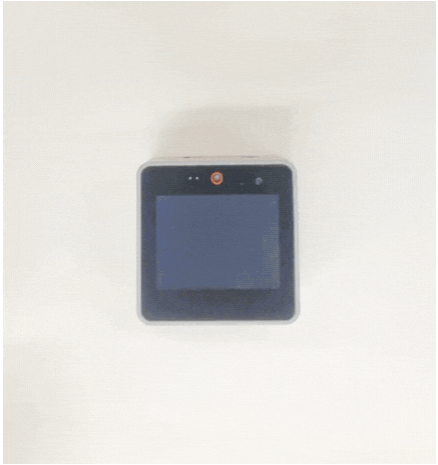
Learn

BMM150 Magnetic Interference

Products containing magnets may interfere with the BMM150 magnetic sensor, resulting in abnormal readings. When used with M5 controllers that have magnets, remove the magnets and avoid placing the BMM150 near strong magnetic fields.

Download Mode

Press and hold the RESET button for 3 seconds. When the green indicator LED turns on, release the button. After release, the green indicator LED turns off, indicating that the device has entered download mode.



Power On/Off

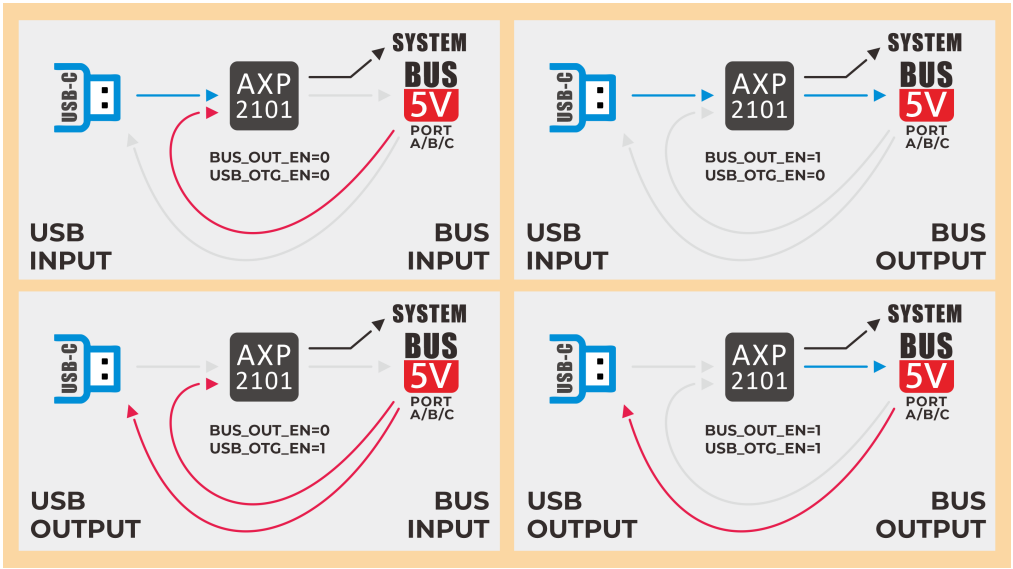
- Power-on: single-click the left POWER button
- Power-off: long-press the left POWER button for 6 s
- Reset: single-click the bottom RST button

IMU Triaxial Direction Schematic Diagram



Power Management

CoreS3 uses the AXP2101 PMU together with the AW9523B IO expander to control power input/output directions. Refer to the pin states of `BUS_OUT_EN` and `USB_OTG_EN` in the figure below and check the example code in CoreS3 Power Manager Example。

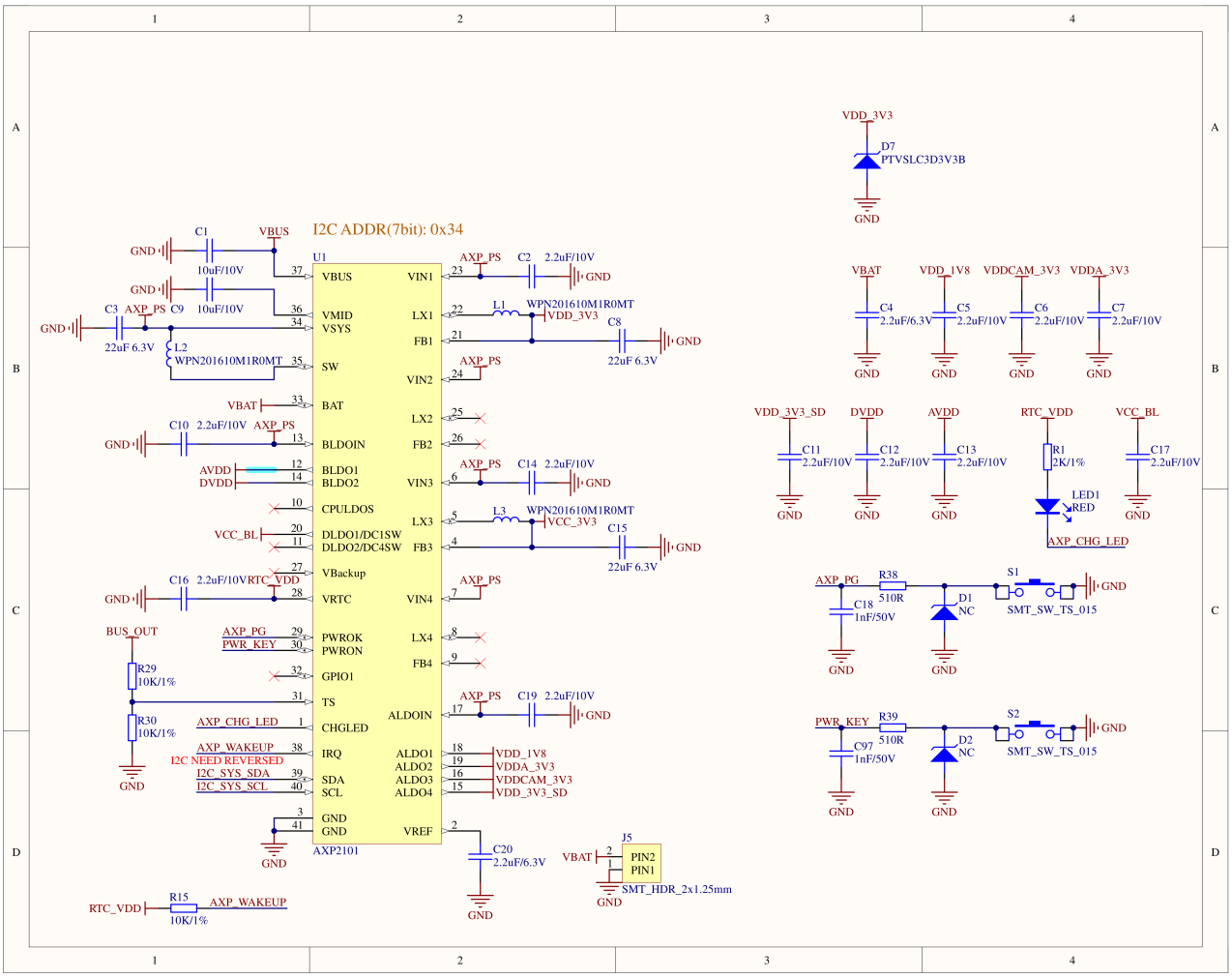


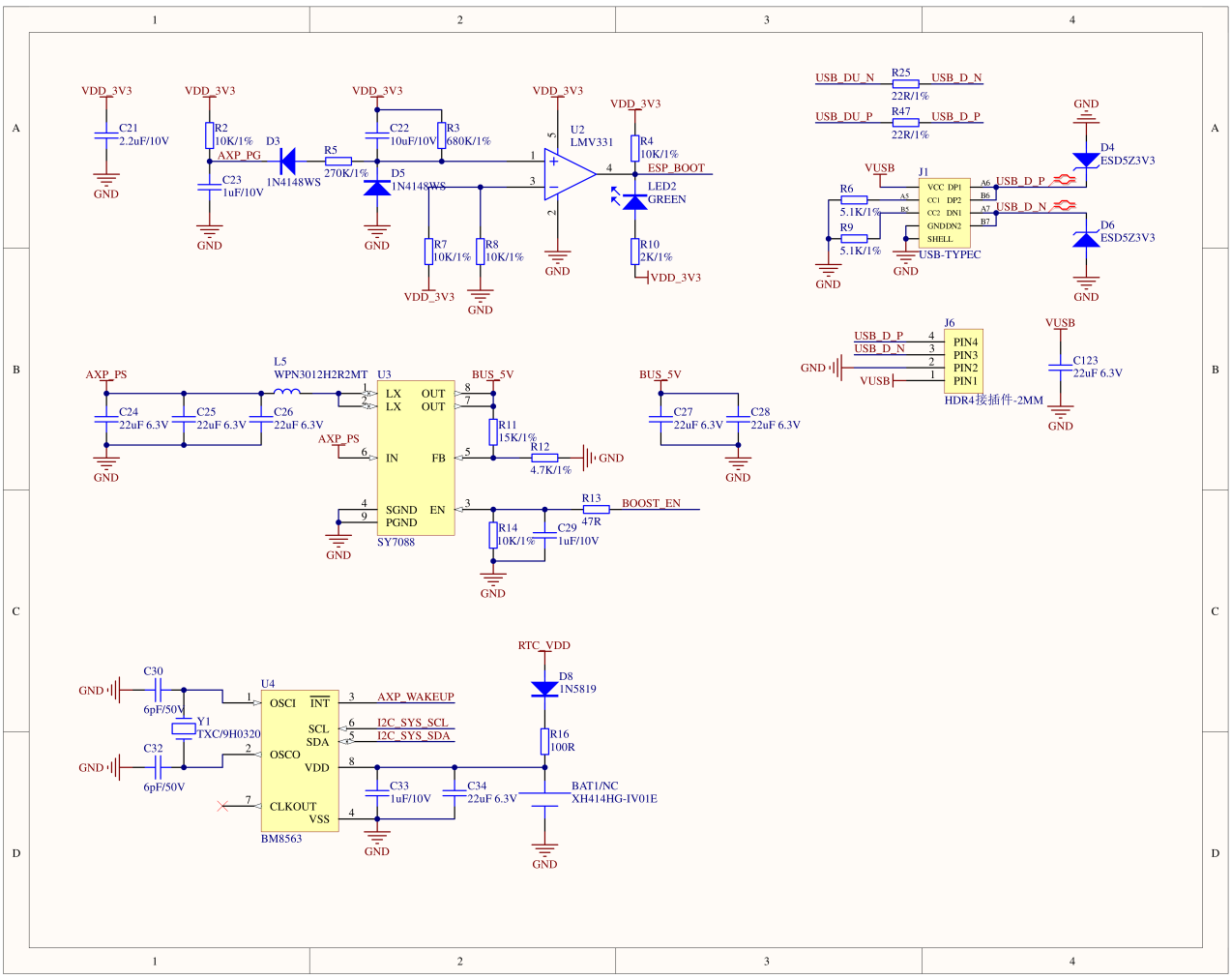
Certifications

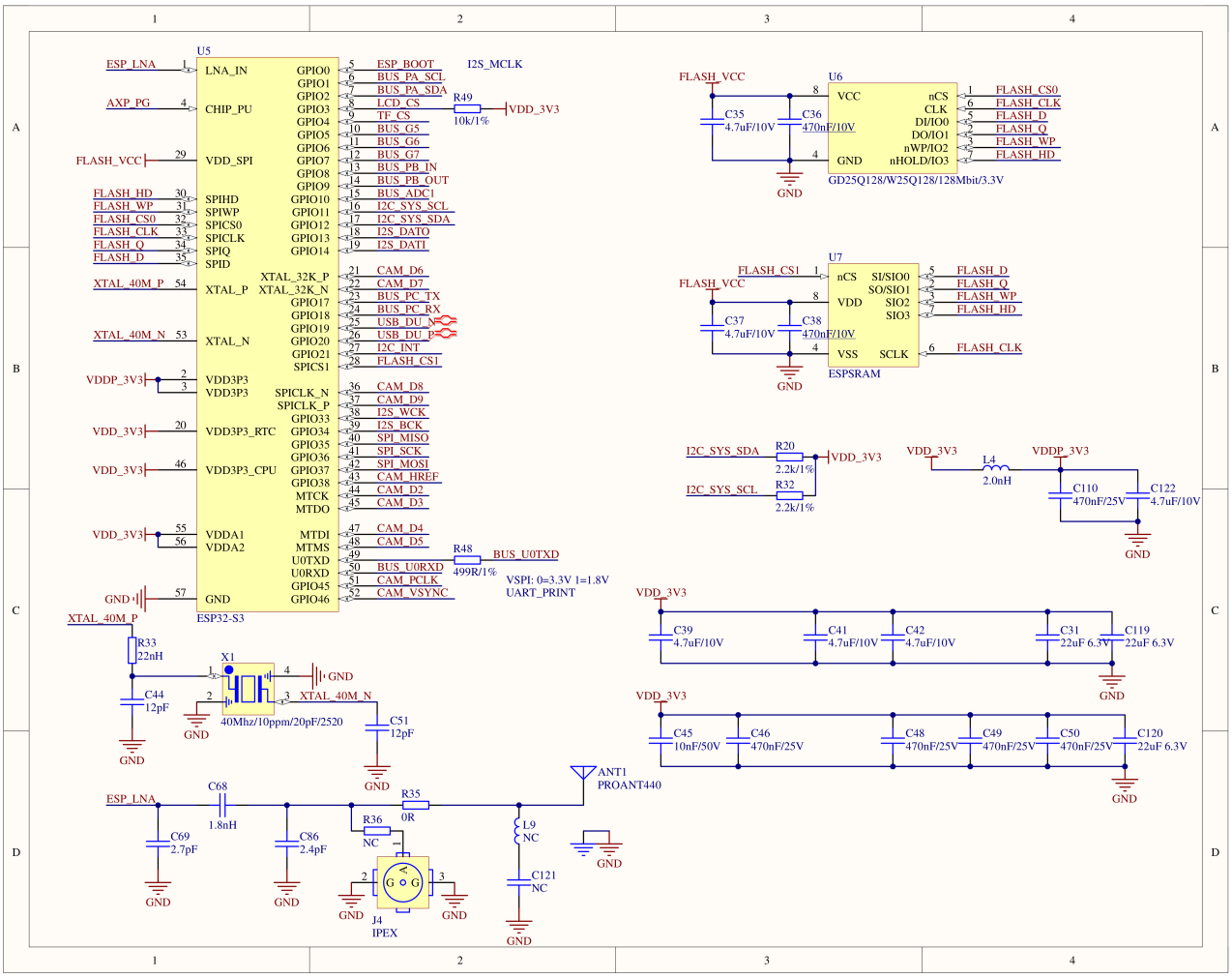
- CE/MIC/FCC/SAR

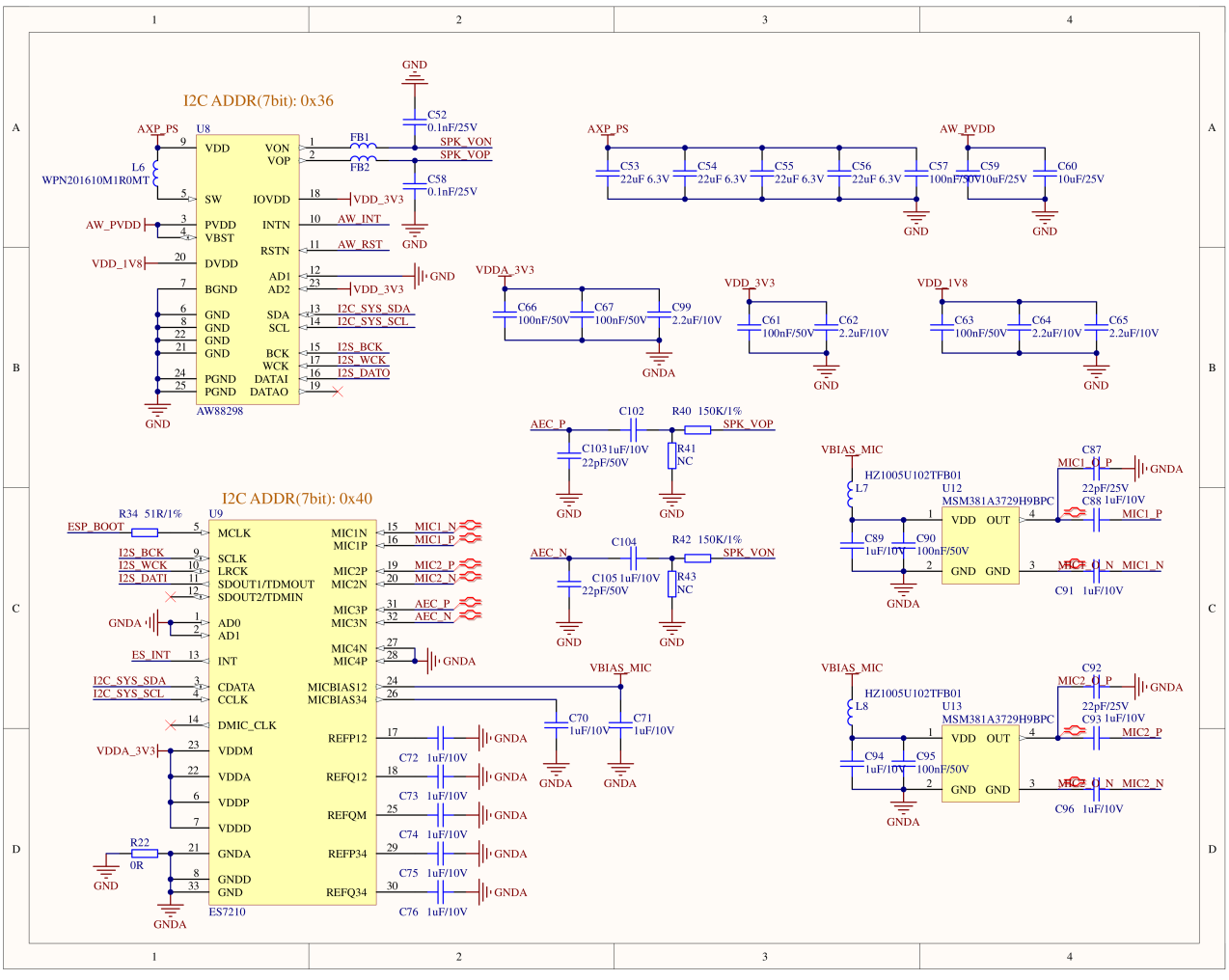
Schematics

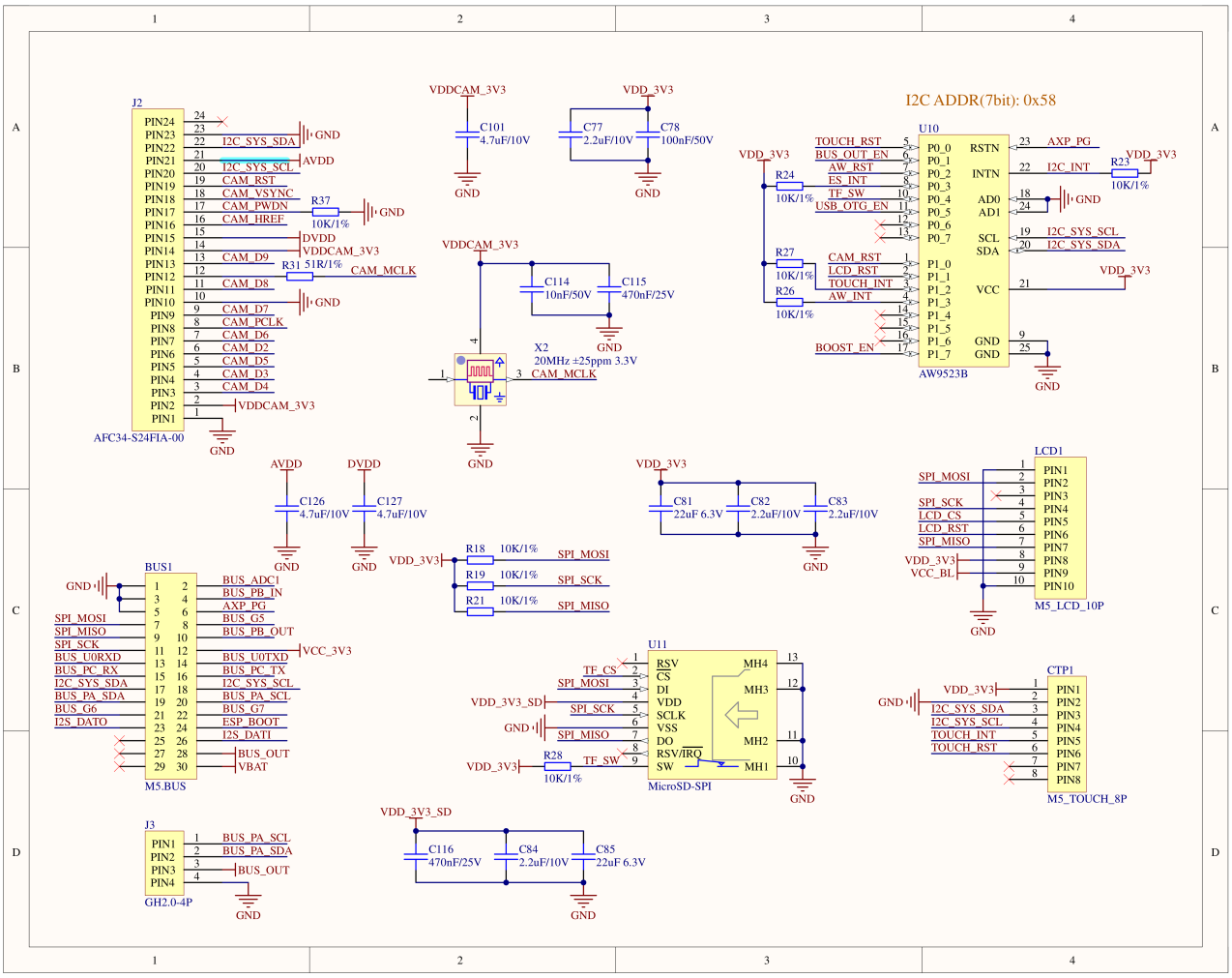
- CoreS3 Schematics PDF
- Base DIN Schematics PDF

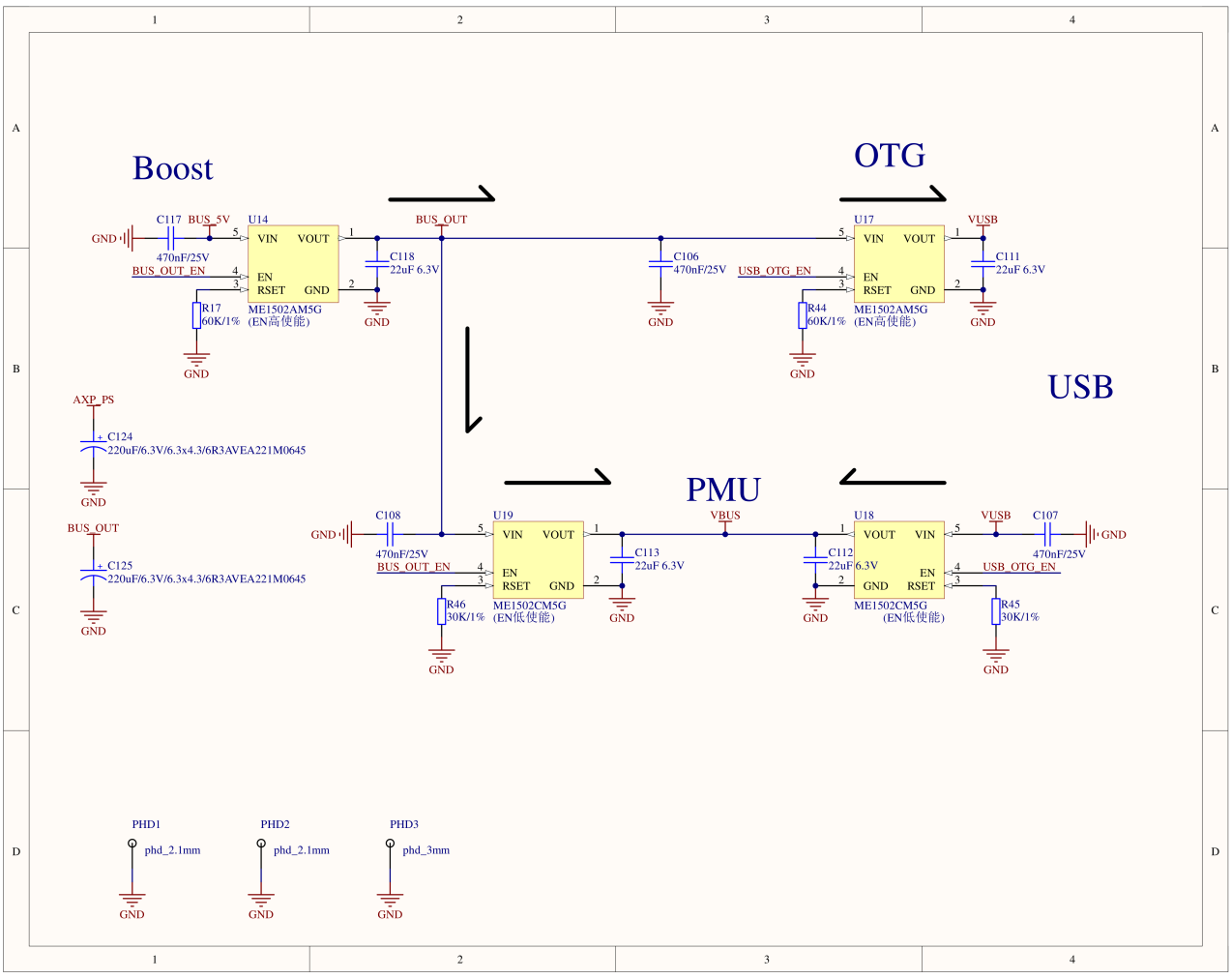

I2C ADDR(7bit): 0x34
U1
AXP2101
VBUS
VMID
VSYS
SW
BAT
BLDOIN
BLDO1
BLDO2
CPULDOS
DLDO1/DC1SW
DLDO2/DC4SW
VBackup
VRTC
PWROK
PWRON
GPIO1
TS
CHGLED
IRQ
SDA
SCL
GND
GND
VIN1
LX1
FB1
VIN2
LX2
FB2
VIN3
LX3
FB3
VIN4
LX4
FB4
ALDOIN
ALDO1
ALDO2
ALDO3
ALDO4
VREF
C1
10uF/10V
C3
22uF 6.3V
C9
10uF/10V
L2
WPN201610M1R0MT
C10
2.2uF/10V
C16
2.2uF/10V
C2
2.2uF/10V
L1
WPN201610M1R0MT
C8
22uF 6.3V
C14
2.2uF/10V
L3
WPN201610M1R0MT
C15
22uF 6.3V
C19
2.2uF/10V
C20
2.2uF/6.3V
R29
10K/1%
R30
10K/1%
R15
10K/1%
AXP_PS
VBAT
AVDD
DVDD
VCC_BL
RTC_VDD
BUS_OUT
AXP_PG
PWR_KEY
AXP_CHG_LED
AXP_WAKEUP
I2C NEED REVERSED
I2C_SYS_SDA
I2C_SYS_SCL
VDD_1V8
VDDA_3V3
VDDCAM_3V3
VDD_3V3_SD
VCC_3V3
J5
PIN2
PIN1
SMT_HDR_2x1.25mm
VDD_3V3
D7
PTVSLC3D3V3B
C4
2.2uF/6.3V
C5
2.2uF/10V
C6
2.2uF/10V
C7
2.2uF/10V
C11
2.2uF/10V
C12
2.2uF/10V
C13
2.2uF/10V
R1
2K/1%
LED1
RED
C17
2.2uF/10V
R38
510R
C18
1nF/50V
D1
NC
S1
SMT_SW_TS_015
R39
510R
C97
1nF/50V
D2
NC
S2
SMT_SW_TS_015
GND

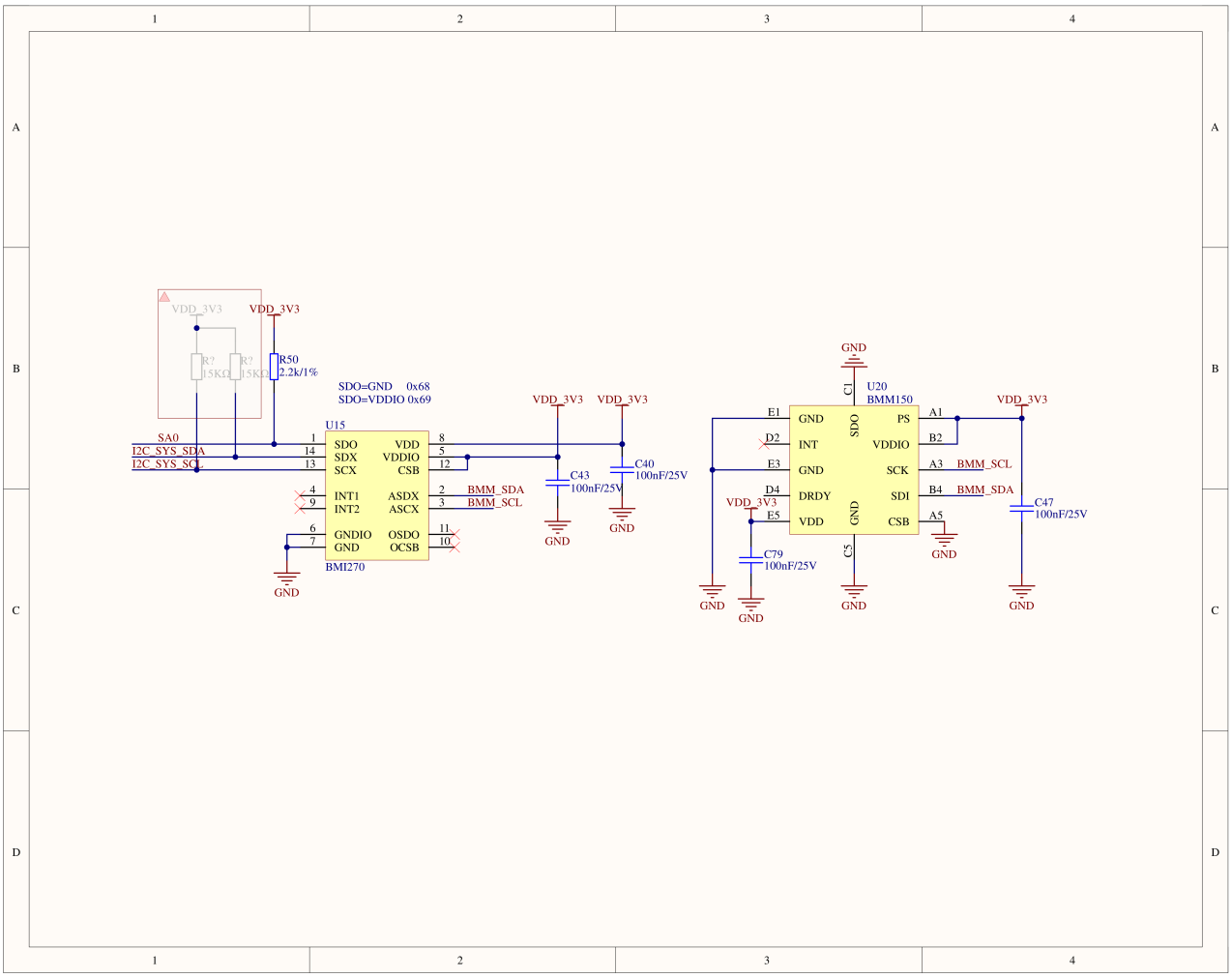

VDD_3V3
C21
2.2uF/10V
R2
10K/1%
D3
1N4148WS
AXP_PG
C23
1uF/10V
R5
270K/1%
C22
10uF/10V
R3
680K/1%
D5
1N4148WS
U2
LMV331
R4
10K/1%
ESP_BOOT
LED2
GREEN
R10
2K/1%
R7
10K/1%
R8
10K/1%
L5
WPN3012H2R2MT
U3
SY7088
LX
OUT
IN
FB
SGND
PGND
EN
BUS_5V
C24
22uF 6.3V
C25
22uF 6.3V
C26
22uF 6.3V
C27
22uF 6.3V
C28
22uF 6.3V
R11
15K/1%
R12
4.7K/1%
R13
47R
R14
10K/1%
C29
1uF/10V
BOOST_EN
USB_DU_N
USB_DU_P
R25
22R/1%
R47
22R/1%
USB_D_N
USB_D_P
VUSB
J1
USB-TYPEC
R6
5.1K/1%
R9
5.1K/1%
D4
ESD5Z3V3
D6
ESD5Z3V3
J6
HDR4接插件-2MM
C123
22uF 6.3V
U4
BM8563
OSCI
OSCO
CLKOUT
INT
SCL
SDA
VDD
VSS
C30
6pF/50V
C32
6pF/50V
Y1
TXC/9H0320
C33
1uF/10V
C34
22uF 6.3V
D8
1N5819
R16
100R
BAT1/NC
XH414HG-IV01E
GND

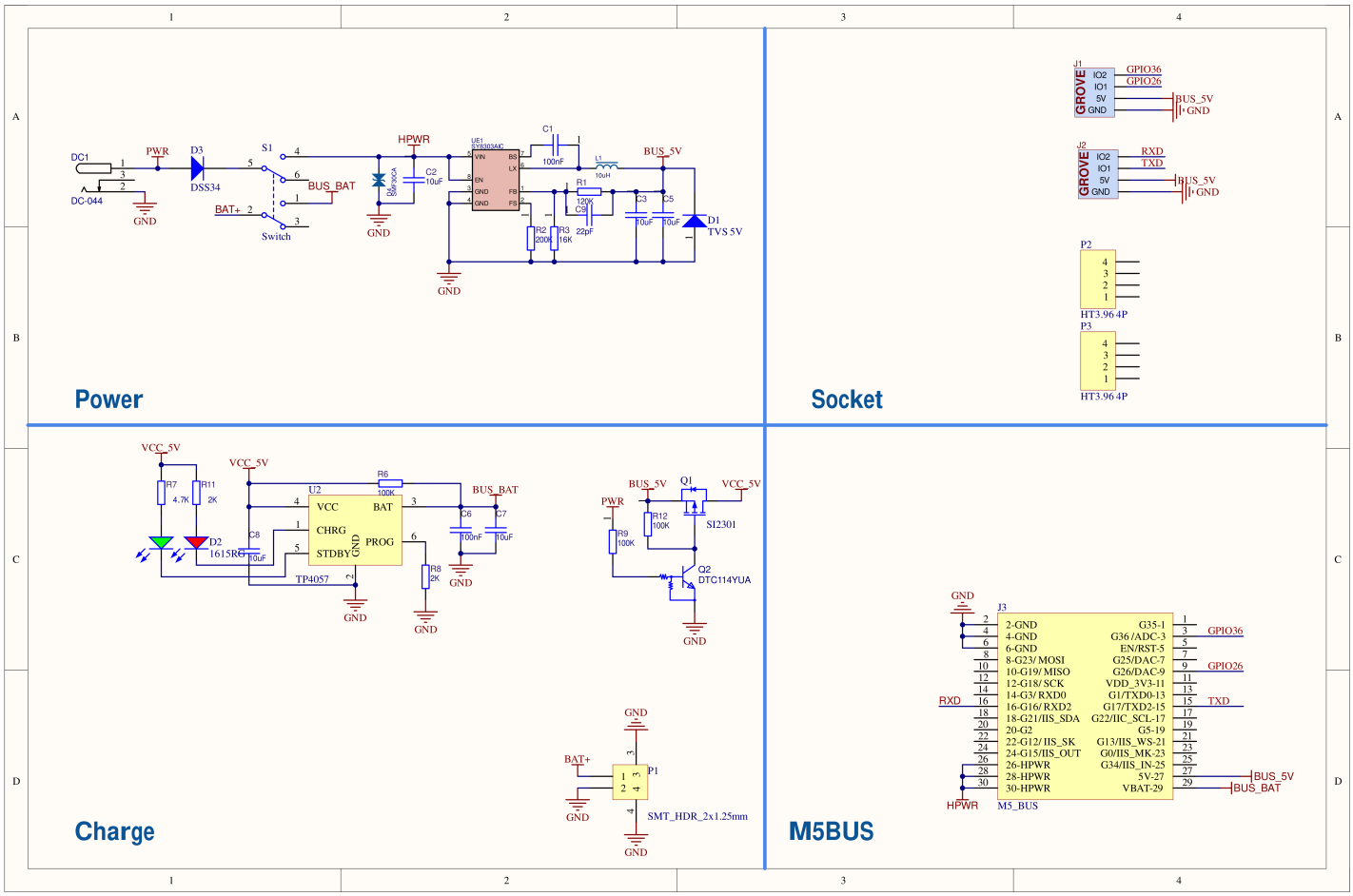

U5
ESP32-S3
LNA_IN
CHIP_PU
VDD_SPI
SPIHD
SPIWP
SPICS0
SPICLK
SPIQ
SPID
XTAL_P
XTAL_N
VDD3P3
VDD3P3_RTC
VDD3P3_CPU
VDDA1
VDDA2
GND
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
XTAL_32K_P
XTAL_32K_N
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
SPICS1
SPICLK_N
SPICLK_P
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
MTCK
MTDO
MTDI
MTMS
U0TXD
U0RXD
GPIO45
GPIO46
ESP_LNA
AXP_PG
FLASH_VCC
FLASH_HD
FLASH_WP
FLASH_CS0
FLASH_CLK
FLASH_Q
FLASH_D
XTAL_40M_P
XTAL_40M_N
VDDP_3V3
VDD_3V3
GND
ESP_BOOT
BUS_PA_SCL
BUS_PA_SDA
LCD_CS
TF_CS
BUS_G5
BUS_G6
BUS_G7
BUS_PB_IN
BUS_PB_OUT
BUS_ADC1
I2C_SYS_SCL
I2C_SYS_SDA
I2S_DATO
I2S_DATI
CAM_D6
CAM_D7
BUS_PC_TX
BUS_PC_RX
USB_DU_N
USB_DU_P
I2C_INT
FLASH_CS1
CAM_D8
CAM_D9
I2S_WCK
I2S_BCK
SPI_MISO
SPI_SCK
SPI_MOSI
CAM_HREF
CAM_D2
CAM_D3
CAM_D4
CAM_D5
BUS_U0RXD
CAM_PCLK
CAM_VSYNC
I2S_MCLK
R49
10k/1%
R48
499R/1%
BUS_U0TXD
VSPI: 0=3.3V 1=1.8V
UART_PRINT
R33
22nH
X1
40M
C44
12pF
40Mhz/10ppm/20pF/2520
C51
12pF
C68
1.8nH
C69
2.7pF
C86
2.4pF
R35
0R
R36
NC
J4
IPEX
L9
NC
C121
NC
ANT1
PROANT440
U6
GD25Q128/W25Q128/128Mbit/3.3V
VCC
GND
nCS
CLK
DI/IO0
DO/IO1
nWP/IO2
nHOLD/IO3
C35
4.7uF/10V
C36
470nF/10V
U7
ESPSRAM
SI/SIO0
SO/SIO1
SIO2
SIO3
VDD
VSS
SCLK
C37
4.7uF/10V
C38
470nF/10V
R20
2.2k/1%
R32
2.2k/1%
L4
2.0nH
C110
470nF/25V
C122
4.7uF/10V
C39
4.7uF/10V
C41
4.7uF/10V
C42
4.7uF/10V
C31
22uF 6.3V
C119
22uF 6.3V
C45
10nF/50V
C46
470nF/25V
C48
470nF/25V
C49
470nF/25V
C50
470nF/25V
C120
22uF 6.3V


I2C ADDR(7bit): 0x36
U8
AW88298
VDD
VON
VOP
SW
IOVDD
PVDD
VBST
INTN
RSTN
DVDD
AD1
AD2
BGND
SDA
SCL
BCK
WCK
DATAI
DATAO
PGND
AXP_PS
AW_PVDD
VDD_1V8
VDD_3V3
L6
WPN201610M1R0MT
FB1
FB2
SPK_VON
SPK_VOP
C52
0.1nF/25V
C58
0.1nF/25V
AW_INT
AW_RST
I2C_SYS_SDA
I2C_SYS_SCL
I2S_BCK
I2S_WCK
I2S_DATO
C53
22uF 6.3V
C54
22uF 6.3V
C55
22uF 6.3V
C56
22uF 6.3V
C57
100nF/50V
C59
10uF/25V
C60
10uF/25V
VDDA_3V3
C66
100nF/50V
C67
100nF/50V
C99
2.2uF/10V
C61
100nF/50V
C62
2.2uF/10V
C63
100nF/50V
C64
2.2uF/10V
C65
2.2uF/10V
GNDA
I2C ADDR(7bit): 0x40
U9
ES7210
MCLK
SCLK
LRCK
SDOUT1/TDMOUT
SDOUT2/TDMIN
AD0
AD1
INT
CDATA
CCLK
DMIC_CLK
VDDM
VDDA
VDDP
VDDD
GNDA
GNDD
GND
MIC1N
MIC1P
MIC2P
MIC2N
MIC3P
MIC3N
MIC4N
MIC4P
MICBIAS12
MICBIAS34
REFP12
REFQ12
REFQM
REFP34
REFQ34
ESP_BOOT
R34 51R/1%
ES_INT
R22
0R
MIC1_N
MIC1_P
MIC2_P
MIC2_N
AEC_P
AEC_N
C102
R40 150K/1%
C103 1uF/10V
22pF/50V
R41
NC
C104
R42 150K/1%
C105 1uF/10V
22pF/50V
R43
NC
VBIAS_MIC
C70
1uF/10V
C71
1uF/10V
C72 1uF/10V
C73 1uF/10V
C74 1uF/10V
C75 1uF/10V
C76 1uF/10V
HZ1005U102TFB01
L7
L8
U12
U13
MSM381A3729H9BPC
C87
C88 1uF/10V
C89
1uF/10V
C90
100nF/50V
C91 1uF/10V
C92
C93 1uF/10V
C94
1uF/10V
C95
100nF/50V
C96 1uF/10V


I2C ADDR(7bit): 0x58
J2
AFC34-S24FIA-00
U10
AW9523B
X2
20MHz ±25ppm 3.3V
LCD1
M5_LCD_10P
BUS1
M5.BUS
U11
MicroSD-SPI
CTP1
M5_TOUCH_8P
J3
GH2.0-4P


Boost
OTG
USB
PMU
U14
ME1502AM5G
(EN高使能)
U17
ME1502AM5G
(EN高使能)
U19
ME1502CM5G
(EN低使能)
U18
ME1502CM5G
(EN低使能)
PHD1
phd_2.1mm
PHD2
phd_2.1mm
PHD3
phd_3mm


VDD_3V3
R50
2.2k/1%
SDO=GND 0x68
SDO=VDDIO 0x69
U15
SA0
I2C_SYS_SDA
I2C_SYS_SCL
SDO
SDX
SCX
VDD
VDDIO
CSB
INT1
INT2
ASDX
ASCX
GNDIO
GND
OSDO
OCSB
BMI270
BMM_SDA
BMM_SCL
C43
100nF/25V
C40
100nF/25V
U20
BMM150
PS
INT
DRDY
SDI
SCK
VDDIO
C79
C47
GND


Power
DC1
DC-044
PWR
D3
DSS34
S1
Switch
BAT+
BUS_BAT
HPWR
C1
C2
L1
10uH
R1
R2
R3
C3
C5
D1
TVS 5V
Socket
J1
J2
GROVE
GPIO36
GPIO26
RXD
TXD
P2
P3
HT3.96 4P
Charge
VCC_5V
R7
R11
R6
U2
VCC
BAT
CHRG
STDBY
PROG
D2
TP4057
C8
C26
C7
R8
Q1
SI2301
R12
R9
Q2
DTC114YUA
P1
SMT_HDR_2x1.25mm
M5BUS
J3
2-GND
4-GND
6-GND
8-G23/MOSI
10-G19/MISO
12-G18/SCK
14-G3/RXD0
16-G16/RXD2
18-G21/IIS_SDA
20-G2
22-G12/IIS_SK
24-G15/IIS_OUT
26-HPWR
28-HPWR
30-HPWR
G35-1
G36/ADC-3
EN/RST-5
G25/DAC-7
G26/DAC-9
VDD_3V3-11
G1/TXD0-13
G17/TXD2-15
G22/IIC_SCL-17
G5-19
G13/IIS_WS-21
G0/IIS_MK-23
G34/IIS_IN-25
5V-27
VBAT-29
M5_BUS

PinMap

LCD Screen

LCD Pixels: 320x240

ESP32-S3	G37	G36	G3	G35
ILI9342C	MOSI	SCK	CS	DC

AW9523B (0x58)	P1_1
ILI9342C	RST

AXP2101 (0x34)	DLDO1	LX1
ILI9342C	BL	PWR

microSD

TF Card Maximum Supported Capacity: 16GB

ESP32-S3	G35	G37	G36	G4
TF Card	MISO	MOSI	SCK	CS

Camera & Proximity Sensor LTR-553ALS-WA

ESP32-S3	G12	G11	G45	G46	G38
GC0308 (0x21)	I2C_SYS_SDA	I2C_SYS_SCL	CAM_PCLK	CAM_VSYNC	CAM_HREF
LTR-553ALS-WA (0x23)	I2C_SYS_SDA	I2C_SYS_SCL			

AW9523B	P1_0
GC0308	CAM_RST

> The LTR-553ALS-WA proximity sensor and the camera are integrated on one ribbon cable, and use I2C for communication. Please refer to the table above for detailed communication addresses.

GC0308

接口	Camera Pin	ESP32-S3
SCCB Clock	SIOC	G11
SCCB Data	SIOD	G12
System Clock	XCLK	-1
Vertical Sync	VSYNC	G46
Horizontal Reference	HREF	G38
Pixel Clock	PCLK	G45
Pixel Data Bit 0	D0	G39
Pixel Data Bit 1	D1	G40
Pixel Data Bit 2	D2	G41
Pixel Data Bit 3	D3	G42
Pixel Data Bit 4	D4	G15
Pixel Data Bit 5	D5	G16
Pixel Data Bit 6	D6	G48
Pixel Data Bit 7	D7	G47
Camera Reset	RESET	-1
Camera Power Down	PWDN	-1

CAP.TOUCH

ESP32-S3	G12	G11
FT6336U (0x38)	I2C_SYS_SDA	I2C_SYS_SCL

AW9523B	P0_0	P1_2
FT6336U	TOUCH_RST	TOUCH_INT

Microphone & Amplifier

ESP32-S3	G12	G11	G34	G33	G13	G14	G0
ES7210 (0x40)	I2C_SYS_SDA	I2C_SYS_SCL	I2S_BCK	I2S_WCK		I2S_DATI	I2S_MCLK
AW88298 (0x36)	I2C_SYS_SDA	I2C_SYS_SCL	I2S_BCK	I2S_WCK	I2S_DATO		

AW9523B	P0_2	P1_3
AW88298	AW_RST	AW_INT

AXP Power Indicator Light

AXP2101	AXP_CHG_LED
Red LED	RTC_VDD

RTC

ESP32-S3	G12	G11
BM8563 (0x51)	I2C_SYS_SDA	I2C_SYS_SCL

AXP2101	IRQ
BM8563	AXP_WAKEUP

IMU (3-axis Gyroscope + 3-axis Accelerometer + 3-axis Magnetometer)

ESP32-S3	G12	G11
BMI270 (0x69)	I2C_SYS_SDA	I2C_SYS_SCL

Internal I2C Connection

ESP32-S3	G12	G11
BMI270	I2C_SYS_SDA	I2C_SYS_SCL
AXP2101	I2C_SYS_SDA	I2C_SYS_SCL
BM8563	I2C_SYS_SDA	I2C_SYS_SCL
ES7210	I2C_SYS_SDA	I2C_SYS_SCL
AW88298	I2C_SYS_SDA	I2C_SYS_SCL

BMM150

BMI270	BMI270_ASDx	BMI270_ASCx
BMM150 (0x10)	BMM_SDA	BMM_SCL

BMM150 Mounted on BMI270

Connected through BMI270's Sensor Hub auxiliary I2C interface to access BMM150 and achieve unified 9-axis sensor data acquisition.

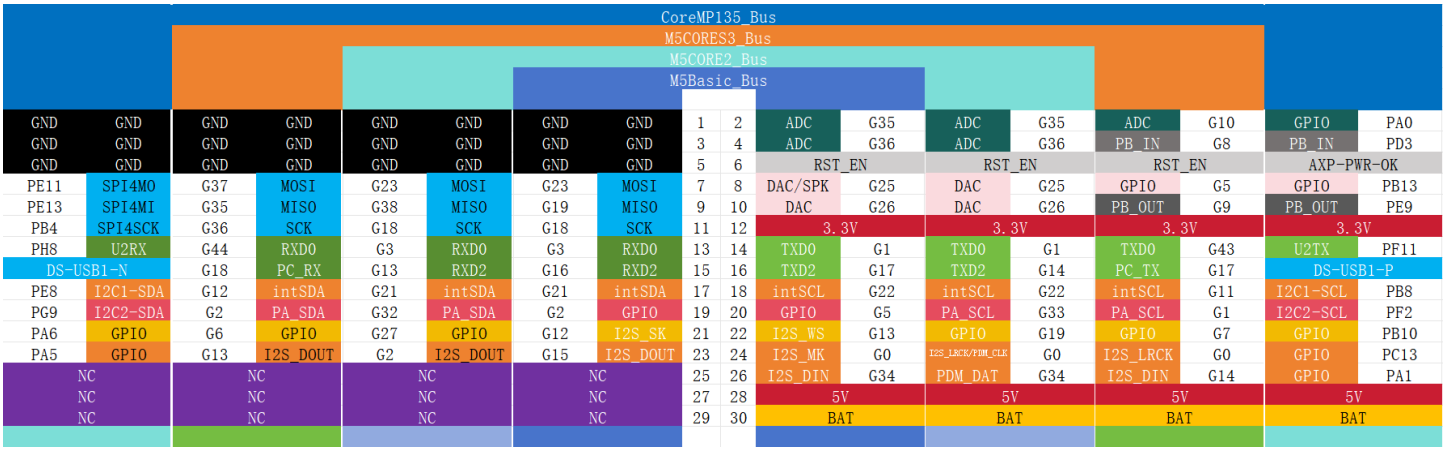
HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	G2	G1
PORT.B	GND	5V	G9	G8
PORT.C	GND	5V	G17	G18

CoreS3 M5-Bus Diagram

FUNC	PIN	LEFT	RIGHT	PIN	FUNC
	GND	1	2	G10	ADC
	GND	3	4	G8	PB_IN
	GND	5	6	RST	EN
MOSI	G37	7	8	G5	GPIO
MISO	G35	9	10	G9	PB_OUT
SCK	G36	11	12	3V3	
RXD0	G44	13	14	G43	TXD0
PC_RX	G18	15	16	G17	PC_TX
Int SDA	G12	17	18	G11	Int SCL
PORT.A SDA	G2	19	20	G1	PORT.A SCL
GPIO	G6	21	22	G7	GPIO
I2S_DOUT	G13	23	24	G0	I2S_LRCK
	HVIN(Base DIN)	25	26	G14	I2S_DIN
	HVIN(Base DIN)	27	28	5V	
	HVIN(Base DIN)	29	30	BAT	

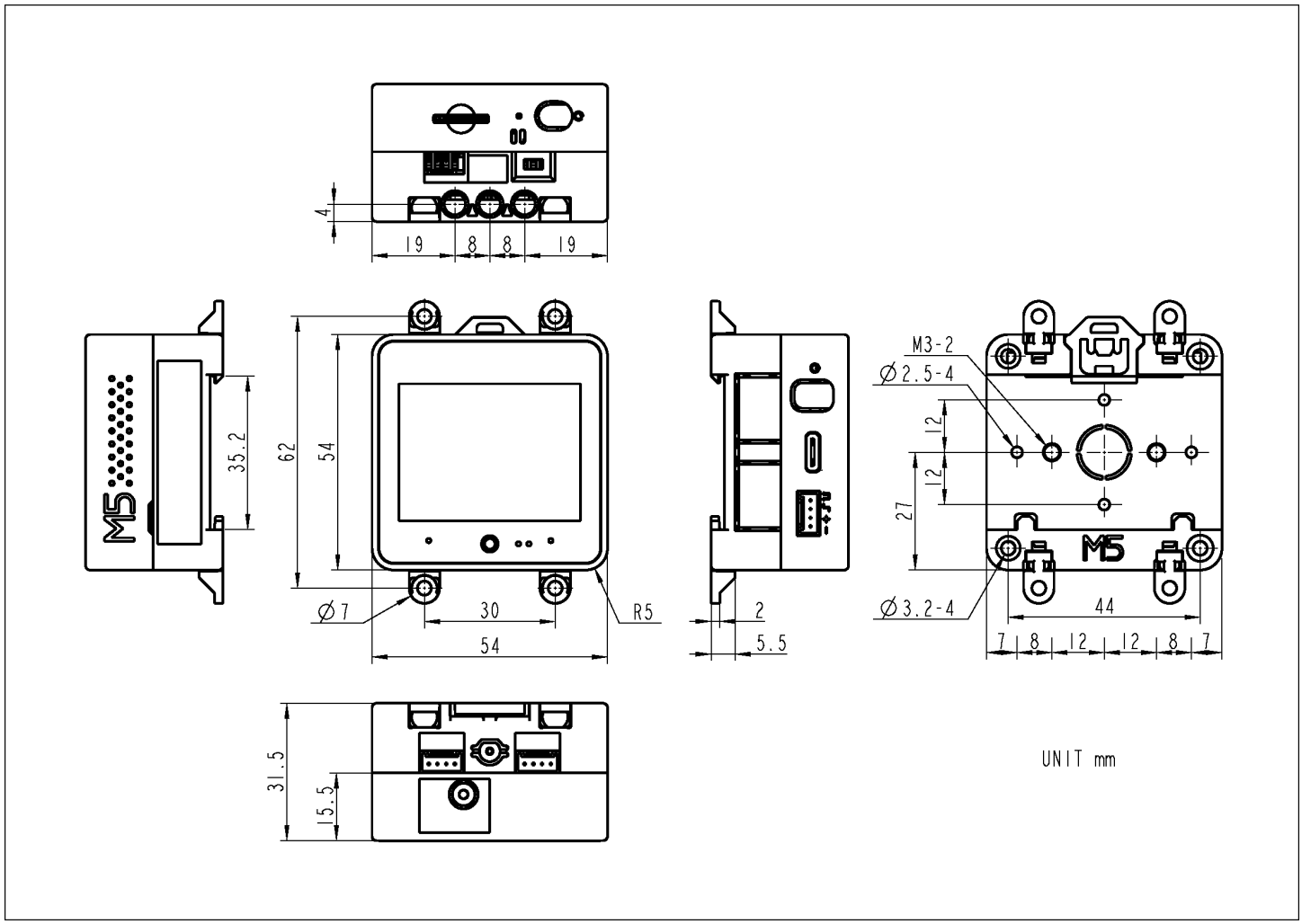
Core Series PinMap Comparison



GND	GND	GND	GND	GND	GND	GND	GND	1	2	ADC	G35	ADC	G35	ADC	G10	GPIO	PA0
GND	GND	GND	GND	GND	GND	GND	GND	3	4	ADC	G36	ADC	G36	PB_IN	G8	PB_IN	PD3
GND	GND	GND	GND	GND	GND	GND	GND	5	6	RST_EN		RST_EN		RST_EN		AXP-PWR-OK	
PE11	SPI4MO	G37	MOSI	G23	MOSI	G23	MOSI	7	8	DAC/SPK	G25	DAC	G25	GPIO	G5	GPIO	PB13
PE13	SPI4MI	G35	MISO	G38	MISO	G19	MISO	9	10	DAC	G26	DAC	G26	PB_OUT	G9	PB_OUT	PE9
PB4	SPI4SCK	G36	SCK	G18	SCK	G18	SCK	11	12	3.3V		3.3V		3.3V		3.3V	
PH8	U2RX	G44	RXD0	G3	RXD0	G3	RXD0	13	14	TXD0	G1	TXD0	G1	TXD0	G43	U2TX	PF11
DS-USB1-N		G18	PC_RX	G13	RXD2	G16	RXD2	15	16	TXD2	G17	TXD2	G14	PC_TX	G17	DS-USB1-P	
PE8	I2C1-SDA	G12	intSDA	G21	intSDA	G21	intSDA	17	18	intSCL	G22	intSCL	G22	intSCL	G11	I2C1-SCL	PB8
PG9	I2C2-SDA	G2	PA_SDA	G32	PA_SDA	G2	GPIO	19	20	GPIO	G5	PA_SCL	G33	PA_SCL	G1	I2C2-SCL	PF2
PA6	GPIO	G6	GPIO	G27	GPIO	G12	I2S_SK	21	22	I2S_WS	G13	GPIO	G19	GPIO	G7	GPIO	PB10
PA5	GPIO	G13	I2S_DOUT	G2	I2S_DOUT	G15	I2S_DOUT	23	24	I2S_MK	G0	I2S_LRCK/PDM_CLK	G0	I2S_LRCK	G0	GPIO	PC13
NC		NC		NC		NC		25	26	I2S_DIN	G34	PDM_DAT	G34	I2S_DIN	G14	GPIO	PA1
NC		NC		NC		NC		27	28	5V		5V		5V		5V	
NC		NC		NC		NC		29	30	BAT		BAT		BAT		BAT	

Model Size

- Cores3 Model Size DXF



PCB

- Cores3 PcbDoc

Description

This file can be used for DIY adaptation of CoreS3 Module Type A expansion modules.

Structure

- CoreS3 Structure Files

Datasheets

- ESP32-S3
- LTR-553ALS-WA
- GC0308
- ES7210
- BMM150
- BMI270
- BM8563
- AXP2101
- AW88298
- AW9523B

Softwares

Quick Start

- CoreS3 OpenAI Voice Assistant
- CoreS3 XiaoZhi Voice Assistant
- CoreS3 HA Voice Assistant
- CoreS3 ESP-Claw Firmware Flashing

Arduino

- CoreS3 Arduino Quick Start
- CoreS3 Arduino Library
- CoreS3 Arduino M5Unified Library
- CoreS3 Arduino M5GFX Library

UiFlow2

- CoreS3 UiFlow2 Quick Start
- CoreS3 UiFlow2 Book

PlatformIO

- CoreS3 Factory Firmware

```
[env:m5stack-cores3]
platform = espressif32@6.7.0
board = esp32-s3-devkitc-1
framework = arduino
upload_speed = 1500000
monitor_speed = 115200
build_flags =
    -DESP32S3
    -DBOARD_HAS_PSRAM
    -mfix-esp32-psram-cache-issue
    -DCORE_DEBUG_LEVEL=5
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DARDUINO_USB_MODE=1
lib_deps =
    M5Unified=https://github.com/m5stack/M5Unified
```

ESP-IDF

- Espressif's Board Support Packages – CoreS3
- CoreS3 ESP-IDF BSP Guide

Easyloader

Easyloader	Download	Note
CoreS3 Factory Firmware Easyloader	download	/

Flash CoreS3 factory firmware with Web Tools

Other

- CoreS3 Restore Factory Firmware Guide
- I2C Address Table

Chip	Address
AW88298	0x36
AW9523	0x58
AXP2101	0x34
BM8563	0x51
BMI270	0x69
BMM150	0x10
ES7210	0x40
FT6336	0x38
GC0308	0x21
LTR553	0x23

Video

Product Comparison

Hardware Peripheral			
	CoreS3-Lite	CoreS3	CoreS3-SE
Camera (GC0308)	✅	✅	❌
Proximity Sensor (LTR-553ALS-WA)	✅	✅	❌
IMU (BMI270)	✅	✅	❌
Compass (BMM150)	✅	✅	❌
RTC	✅	✅	✅
Microphone	✅	✅	✅
Speaker	✅	✅	✅
PMIC (AXP2101)	✅	✅	✅
16 MB Flash & 8 MB PSRAM	✅	✅	✅
Touch	✅	✅	✅
Back Cover	Cover For CoreS3	Base DIN	❌
Battery Capacity	200 mAh	500 mAh	❌

To compare information on the controller series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.